

# Fujitsu

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## 1 Fujitsu formulations

### Fixed order book embedding

we are given as input a graph  $G = (V, E)$  a vertex ordering  $\prec$ , and a fixed number of pages  $k$ . We aim to find an assignment of the edges to the pages such that the number of crossing on each page is minimized.

$$x_{e,p} = \begin{cases} 1, & \text{if edge } e \text{ is assigned to page } p \\ 0, & \text{otherwise} \end{cases}$$

We can precompute the edge crossing pairs and store them in  $\mathcal{C}$ , two edges  $e = (u, v)$  and  $f = (x, y)$  crosses if  $u \prec x \prec v \prec y$  (or simmetries).

we need to enforce that each edge is assigned to exactly one page:

$$\sum_{p=1}^k x_{e,p} = 1, \forall e \in E$$

as a penalty term

$$H_{page} = A \sum_{e \in E} \left(1 - \sum_{p=1}^k x_{e,p}\right)^2$$

Crossing cost

$$H_{cross} B \sum_{(e,f) \in \mathcal{C}} \sum_{p=1}^k x_{e,p} x_{f,p}$$

final hamiltonian

$$H = H_{page} + H_{cross}$$

### Fixed-Order Skewness Problem

GENERAL SETTING

Let  $G = (V, E)$  be a graph with fixed vertices position (in plane or line) and let  $\mathcal{C}$  be the set of crossing edge pairs in such a configuration.